Stochastic Processes Semester 1

Tutorial 6 – Solutions

1. Toss a coin independently $n_1 + n_2 + n_3$ times, then the number of heads will be $X_1 + X_2 + X_3$ and so

$$X_1 + X_2 + X_3 \sim B(n_1 + n_2 + n_3, p).$$

2. Let us define $\lambda_n = np_n$. Then by assumption $\lambda_n \to 0$ as $n \to \infty$. Further,

$$1 \geq \mathbf{P}(X_n = 0) = (1 - p_n)^n = (1 - \lambda_n/n)^n.$$

For any $\varepsilon > 0$, choose n_0 such that $\lambda_n \leq \varepsilon$ for all $n \geq n_0$. Then, for $n \geq n_0$,

$$(1 - \lambda_n/n)^n \geq (1 - \varepsilon/n)^n \to e^{-\varepsilon}.$$

Thus,

$$\liminf_{n\to\infty} \mathbf{P}(X_n = 0) \geq e^{-\varepsilon}$$

for any $\varepsilon > 0$. Now let ε to 0. Then

$$\liminf_{n\to\infty} \mathbf{P}(X_n = 0) \geq 1.$$

But any probability is ≤ 1. Therefore, we get the desired result.

3. Consider the 2-step procedure and then use Property Π3. The procedure is:
(1) denote $q+r = u$. Then colour each ball "red" with probability p and, say, "beige", with probability u. From Property Π3, we get two independent r.v.'s. $R \sim \Pi(p\lambda)$ and $B \sim \Pi(u\lambda)$. By recolouring differently we arrive at (i)
(2) Now: recolour each beige ball "white" with probability q/u and "yellow" with probability r/u and use the same Property Π3. So

$$\begin{aligned}
\mathbf{P}(R = k; W = i; Y = j) &= \mathbf{P}(R = k; W = i; B = i + j) \\
&= \mathbf{P}(R = k | W = i; B = i + j)\mathbf{P}(W = i; B = i + j) \\
&= \mathbf{P}(R = k)\mathbf{P}(W = i; Y = j) \\
&= \mathbf{P}(R = k)\mathbf{P}(W = i)\mathbf{P}(Y = j),
\end{aligned}$$

where the third equality follows by the Property Π3 for R and B, and the fourth equality follows from applying the splitting theorem to B.

4. For any two events A and B of positive probability,

$$\mathbf{P}(A \mid B) = \frac{\mathbf{P}(A \cap B)}{\mathbf{P}(B)}.$$

Therefore

$$\begin{aligned}
\mathbf{P}(X + 1 = k \mid X_1 + X_2 + X_3 = l) &= \frac{\mathbf{P}(X_1 = k, X_1 + X_2 + X_3 = l)}{\mathbf{P}(X_1 + X_2 + X_3 = l)} \\
&= \frac{\mathbf{P}(X_1 = k) \cdot \mathbf{P}(X_2 + X_3 = l - k)}{\mathbf{P}(X_1 + X_2 + X_3 = l)}
\end{aligned}$$

due to the independence of the r.v. X_1 from the r.v.'s X_2 and X_3.

Now $X_2 + X_3 = k$ is the event that we toss a coin with probability p of a head k times and we end up with 2 heads, the 2nd on the kth toss. This means that in the previous $k-1$ tosses there must have been 1 head and $k-2$ tails, but we don't know when that 1 head was and so

$$\mathbf{P}(X_2 + X_3 = k) = \sum_{i=1}^{k-1} q^{i-1}pq^{k-i-1}p = \sum_{i=1}^{k-1} q^{k-2}p^2 = (k-1)q^{k-2}p^2.$$

Similarly, $\mathbf{P}(X_1 + X_2 + X_3 = k) = \frac{(k-1)(k-2)}{2}q^{k-3}p^3$. Therefore

$$\mathbf{P}(X_1 = k) = q^{k-1}p, \qquad \mathbf{P}(X_2 + X_3 = l - k) = (l-k-1)q^{l-k-2}p^2$$

and

$$\mathbf{P}(X_1 + X_2 + X_3 = l) = \frac{(l-1)(l-2)}{2}q^{l-3}p^3.$$

Substituting these into the above equation gives

$$\mathbf{P}(X + 1 = k \mid X_1 + X_2 + X_3 = l) = \frac{2(l-k-1)}{(l-1)(l-2)}.$$

Observe that this final result does not depend on p.

5. (i) Since Y_1 and Y_2 are independent,

$$\mathbf{P}(Z \le x) = \mathbf{P}(Y_1 \le x, Y_2 \le x) = \mathbf{P}(Y_1 \le x)\mathbf{P}(Y_2 \le x) = (1-e^{-\alpha_1 x})(1-e^{-\alpha_2 x}).$$

(ii)

$$\begin{aligned}
\mathbf{P}(T \le t) &= \mathbf{P}(T \le t,\, Y_2 > Y_1) + \mathbf{P}(T \le t,\, Y_2 \le Y_1) \\
&= \int_0^\infty f_{Y_1}(y)\mathbf{P}(Y_2 - y \le t \mid Y_2 > y)\mathbf{P}(Y_2 > Y_1 \mid Y_1 = y)dy \\
&+ \int_0^\infty f_{Y_2}(y)\mathbf{P}(Y_1 - y \le t \mid Y_1 > y)\mathbf{P}(Y_1 > Y_2 \mid Y_2 = y)dy \\
&= \int_0^\infty \alpha_1 e^{-\alpha_1 y}(1-e^{-\alpha_2 t})e^{-\alpha_2 y}dy + \int_0^\infty \alpha_2 e^{-\alpha_2 y}(1-e^{-\alpha_1 t})e^{-\alpha_1 y}dy \\
&= \frac{1}{\alpha_1+\alpha_2}\left(\alpha_1(1-e^{-\alpha_2 t}) + \alpha_2(1-e^{-\alpha_1 t})\right).
\end{aligned}$$

Alternatively,

$$\begin{aligned}
\mathbf{P}(T > t) &= \int_0^\infty f_{Y_1}(y)\mathbf{P}(Y_2 > y + t)dy + \int_0^\infty f_{Y_2}(y)\mathbf{P}(Y_1 > y + t)dy \\
&= \int_0^\infty \alpha_1 e^{-\alpha_1 y}e^{-\alpha_2(y+t)}dy + \int_0^\infty \alpha_2 e^{-\alpha_2 y}e^{-\alpha_1(y+t)}dy \\
&= \frac{1}{\alpha_1+\alpha_2}\left(\alpha_1 e^{-\alpha_2 t} + \alpha_2 e^{-\alpha_1 t}\right).
\end{aligned}$$

We then find $\mathbf{P}(T \le t) = 1 - \mathbf{P}(T > t)$ as above.

6. First we have the following fact: if a random variable Y has an exponential distribution with parameter α, then $\mathbf{E}(Y) = 1/\alpha$ and $\mathbf{V}ar(Y) = 1/\alpha^2$.

Now making use of Property $\Gamma 1$, we have that if $Y_1, \ldots, Y_n$ are i.i.d. $Exp(\alpha)$ random variables, then

$$X = Y_1 + \cdots + Y_n \sim \Gamma(\alpha, n).$$

So,

$$\mathbf{E}(X) = \mathbf{E}(Y_1 + \cdots + Y_n) = \mathbf{E}(Y_1) + \cdots + \mathbf{E}(Y_n) = \frac{n}{\alpha}$$

and

$$\mathbf{V}ar(X) = \mathbf{V}ar(Y_1 + \cdots + Y_n) = \mathbf{V}ar(Y_1) + \cdots + \mathbf{V}ar(Y_n) = \frac{n}{\alpha^2}.$$

7. Let $N(\cdot)$ denote the Poisson process of call arrivals and let T_{12} be the time of the twelfth call.

   (a) Then the probability that the $12th$ call arrives before 2 minutes has elapsed is the same as the probability that in a 2 minute period there are at least 12 calls and since $N[0,2]$ is Poisson(8) we have

$$\mathbf{P}(T_{12} \leq 2) = \mathbf{P}(N[0,2] \geq 12) = 0.1119.$$

   (b) In a similar way, we find

$$\begin{aligned} \mathbf{P}(3 < T_{12} \leq 5) &= \mathbf{P}(T_{12} \leq 5) - \mathbf{P}(T_{12} \leq 3) \\ &= \mathbf{P}(N[0,5] \geq 12) - \mathbf{P}(N[0,3] \geq 12) \\ &= 0.9786 - 0.5384 = 0.4402. \end{aligned}$$

8. (a) We have $E(N[0,t]) = \text{var}(N[0,t]) = \lambda t$ and so the required coefficient of variation is

$$\frac{\sqrt{\lambda t}}{\lambda t} = \frac{1}{\sqrt{\lambda t}}.$$

   Note that this is decreasing in t.

   (b) Conditional on the event $N[0,t+s] = n$, the distribution of these n points in $[0,t+s]$ is the same as that of n points uniformly distributed in this interval. Hence the conditional distribution of $N[0,t]$ is binomial with parameters n and $t/(t+s)$.

9. Let the random variable N denote the number of number of passengers arriving in $[0,s]$, and let T denote the total time waited by these passengers. Conditional on the event $N = n$, the distribution of the corresponding n arrival times is the same as that of n times uniformly distributed in $[0,s]$. It follows that the conditional distribution of the time the passengers have to wait is also uniformly distributed in $[0,s]$ and since the expected value of a single $U(0,s)$ random variable is $s/2$, we have

$$E(T|N=n) = n\frac{s}{2} = \frac{ns}{2}.$$

   Thus

$$E(T) = E(E(T|N)) = E\left(\frac{Ns}{2}\right) = \frac{\lambda s^2}{2}.$$